Finite Element Modeling and Analysis of a Thin Plate with a Hole - Patran/Nastran

TP 2 - Finite Element Method Lab Report

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ABSTRACT

This study presents a linear static analysis of a thin aluminum plate featuring a central hole to evaluate stress concentration effects under tensile loading. Using MSC Patran and Nastran, a finite element model was developed with a refined Quad 4 Isomesh and symmetry boundary conditions at $x = L/2$ and $y = L/2$ to optimize computational efficiency and prevent rigid body motion. The numerical results identify the maximum Von Mises stress at the hole boundary, demonstrating the characteristic stress distribution of the structure and validating the influence of mesh density on the accuracy of the stress concentration factor compared to analytical solutions.

Key words. Finite Element Method — Stress Concentration — Patran/Nastran — Von Mises

1. Introduction

In the field of aeronautical engineering, plates with holes are common structural elements, found in fuselage skins, wing ribs, and various brackets. These openings are necessary for weight reduction, access, or fasteners, but they significantly alter the internal stress field.

This report details the work performed during the second FEM lab session. The study focuses on a thin rectangular plate featuring a central circular hole. We utilize the power of the Patran/Nastran suite to simulate a tensile test.

2. Modeling and Results: Study 1

This first study focuses on the initial setup of the finite element model and the preliminary stress analysis of the plate.

2.1. *Geometry Creation*

The geometry was generated by defining key points to outline the plate's boundaries and the central hole, following the dimensions of the exercise :

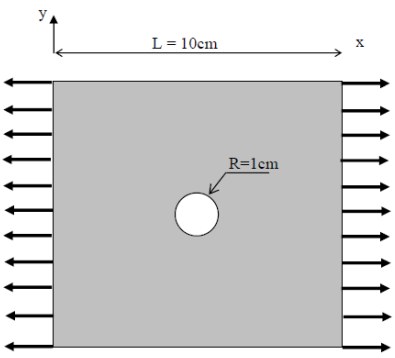

Fig. 1. Dimensions of the plate & hole

We start by creating the corners and points between them, then we link them using curves to creates the edges and the diagonals. Then, we create a 2D circle, and we use the tool "intersect" to create point at the intersection of the curves and the circle; then we delete the center lines inside the circle.

After that, we create the 8 surfaces between the edges, and we obtain this geometry :

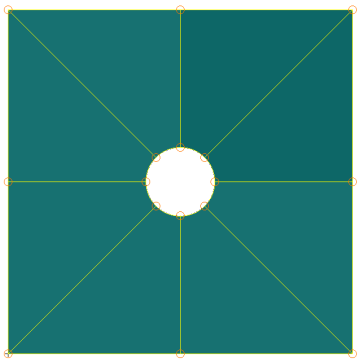

Fig. 2. Geometry of the plate with hole in Patran

2.2. *Meshing*

An **Isomesh** approach was utilized with **Quad 4** elements. A global mesh seed was applied to the surfaces to ensure a structured distribution of elements. We put 5 elements on the edges element, 5 element on the opposite arc of circle, and 8 element on the diagonal elements.

Then, by selecting two internal opposite edges (arc of circle and opposite square edge), we can create the mesh (Fig. 3).

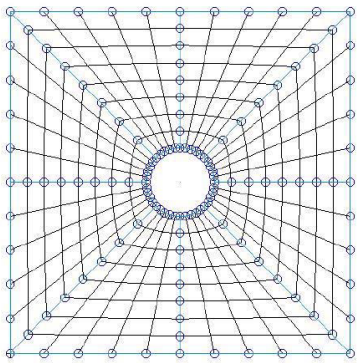

Fig. 3. Plate mesh using quad elements

2.3. *Materials*

The structure is composed of steel, an isotropic material defined by its elastic and strength properties. For the numerical simulation in MSC Patran, the following mechanical characteristics were implemented:

- **Density** (ρ)**:** 7800 kg/m^3
- **Young's Modulus** (E)**:** 2.1×10^{11} Pa (or 210 GPa)
- **Poisson's Ratio** (ν)**:** 0.33
- **Yield Strength** ($\sigma_{R0.2}$)**:** 660 MPa
- **Ultimate Tensile Strength** (σ_{Rm})**:** 849 MPa

Then we assigned the materials to the surfaces/edges using "properties".

2.4. *Loads and Boundary Conditions (BCs)*

2.4.1. First proposition:

First, in order to fix these translations, we create a nodal displacement, and in "input data" we put $< T1, T2, T3 >=< 0,,0 >$ in order to bloc the X and Z translation (translation in Y remains free) called "$x - z$"; and $< T1, T2, T3 >=<, 0, 0 >$ in order to bloc the Y and Z translation (translation in X remains free) called "$y - z$" (we can also directly click on the selected axis to fix the translations).

We can then directly apply 500N to the corners and 1000N to the nodes :

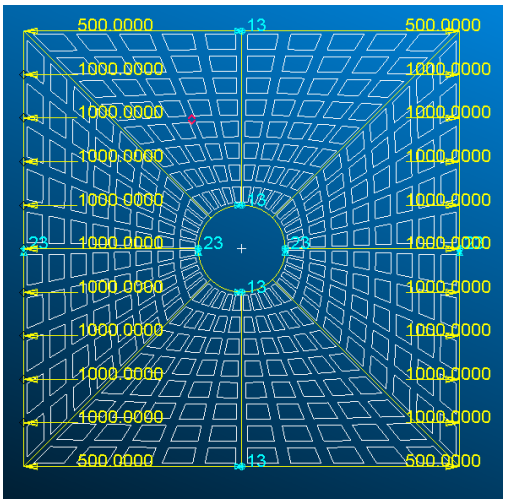

Fig. 4. Forces applied to the nodes

We can indeed see that we have a blocked translation T_2 when $x = L/2$, and a blocked translation T_1 when $y = L/2$; we can also observe the distributed loads.

2.4.2. Second proposition (optional)

We can also directly put distributed tensile loads (1000 N and 500 N) to the corresponding edges. We should select "FEM" option, not "geometry", in order to select the corresponding edges. We can now add $< 100000., 0., 0. >$ in the "edge distributed load" after selecting "CID Distributed Load" for the central edges, and $< 50000., 0., 0. >$ for the corner edges; because CID Distributed Load is taking N/m values as input, not N, and we can calculate these values as follows :

$$\frac{8000N}{0.08m} = 100000N; \frac{1000N}{0.02m} = 50000N$$

But we will choose to continue with the first proposition.

2.5. *Load Cases*

A specific **Load Case** was defined to group the boundary conditions and the applied loads for the first study with the first mesh. So we put all the 4 distributed loads and the symmetries.

Then, it is important to mention the output request. Here, we are going to choose Displacements, Stress, SPCForce and Force.

2.6. *Analysis & Results*

The model was exported to the **MSC Nastran** solver for a Linear Static analysis (**SOL 101**).

After the computation, the results were accessed through the **Results** menu. We focus on the Von Mises stress distribution to identify the areas of maximum stress concentration.

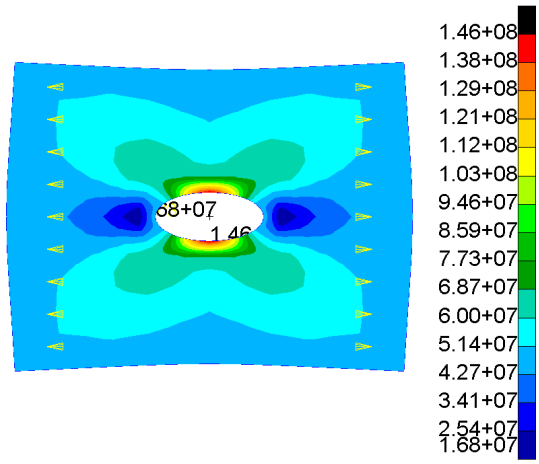

Fig. 5. Von Mises stress distribution around the hole - Case 1

We can also observe the translational displacements :

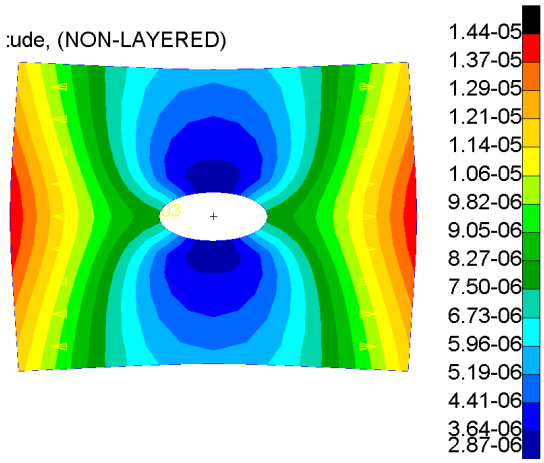

Fig. 6. Translational displacements

We can observe that the maximum Von Mises stress is indeed at an 90° angle from the strain constraint, and the value obtained is 146 MPa, which is significantly lower than the yield strength of the steel ($\sigma_{R0.2}$ = 660 MPa).

3. Modeling and Results: Study 2

In this section, we investigate the influence of the mesh density on the accuracy of the results.

3.1. Mesh Refinement Strategy

The only parameter modified from Study 1 is the **Mesh Seed**. The number of elements was increased, we doubled the number of elements (from 5 to 10 on the edges of the square, and from 8 to 16 in the diagonals); and we obtained the following Von Mises stress distribution values :

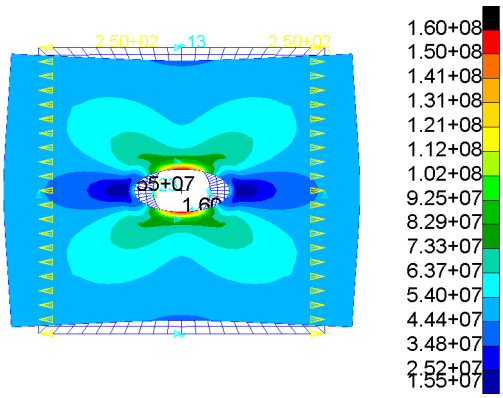

Fig. 7. Von Mises stress distribution around the hole - Case 2

3.2. Impact on Stress Results

The refinement leads to a more precise definition of the stress contours. The maximum stress value observed is different from Study 1, indeed as we can see, we obtained a maximum stress value of 160 MPa, and after calculating the theoritical value of the Von Mises maximum stress, we will confirm that the mesh density is a critical factor in FEA convergence.

4. Questions and Discussion

4.1. Analysis of Study 1 (Question 1)

Maximum Stress: The maximum stress recorded at the opening is 148 MPa.

Theoretical Comparison: According to Peterson's stress concentration charts, the analytical maximum stress is calculated using the gross nominal stress σ_0 and the geometric stress concentration factor K_{tg}:

The nominal stress σ_0 is calculated far from the hole using the gross cross-sectional area (S):

$$S = W \times e = 0.1 \text{ m} \times 0.002 \text{ m} = 0.0002 \text{ m}^2$$

$$\sigma_0 = \frac{F_{tot}}{S} = \frac{10000}{0.0002} = 50 \text{ MPa}$$

For a hole of a diameter $D = 0.02$ m and a plate width $W = 0.1$ m, the ratio is $D/W = 0.2$. According to Peterson's charts, this corresponds to a gross stress concentration factor:

$$K_{tg} \approx 3.14$$

The theoretical maximum stress at the hole boundary is:

$$\sigma_{max}^{th} = K_{tg} \cdot \sigma_0 = 3.14 \times 50 \text{ MPa} = \mathbf{157} \text{ MPa}$$

Interpretation: In case 1, with a non precise mesh seed, we find a **7% difference** between theory and experimental values.

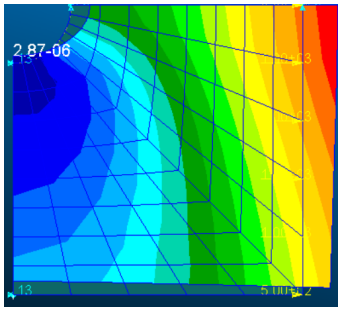

Fig. 9. Translational displacements

4.2. Analysis of Study 2 (Question 2)

- **Detailed Mesh:** The chosen mesh for Study 2 consists of 1281 elements (doubled the number of nodes per segment).
- **Maximum Stress:** With this refined mesh, the peak stress reached 160 MPa.
- **Evolution Analysis:** The evolution of the stress value compared to the previous simulation is due to *mesh convergence*. A finer mesh more accurately represents the true mathematical gradient at the discontinuity. We reached a **≈1.9% difference** with case 2.

5. Modeling and Results: Study 3

Now, we will work with only one quarter of the plate (because of structure and load symmetry). The steps are the same as the previous two cases, so we will directly focus on analyzing the results (with the same number of elements than case 1).

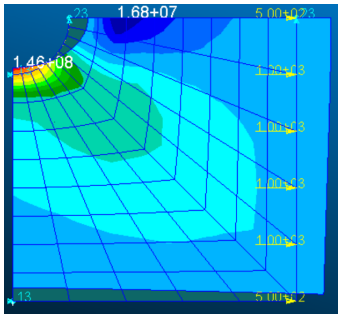

Fig. 8. Stress tensor (Von Mises)

As we can see, we also have a maximum Von Mises constraint of $\sigma_{max}^{quarter} = \mathbf{157}$ MPa $= \sigma_{max}^{case1}$, same as the case 1, so we could only study the quarter of the plate to reduce computational cost and time.

6. Conclusion

This study validates the effectiveness of the finite element method in predicting stress concentration factors for aerospace components. By comparing three modeling approaches, we demonstrated that mesh refinement and the strategic use of symmetry boundary conditions are essential for achieving numerical convergence. The strong correlation between the 160MPa FEM result and Peterson's theoretical value (157MPa) confirms that a well-structured Quad 4 mesh accurately captures local gradients while optimizing computational resources.

References

Larbi, W. (2026). *TP2: Calcul Linéaire d'une Plaque Mince Trouée*. IPSA Course Material.